

The Power of Vaccination

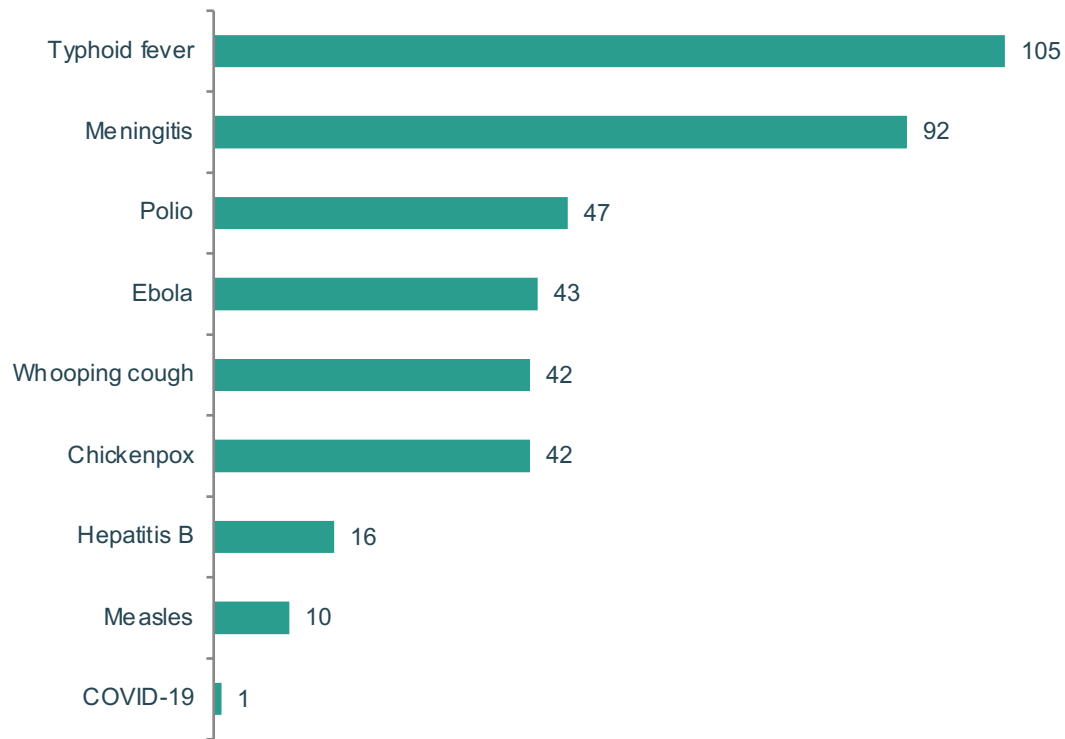
From Smallpox Eradication to COVID-19
in Record Time

1796 – 2020

Source: Our World in Data | ourworldindata.org/vaccination



From 100 Years to Under 1 Year: Vaccine Development Is Accelerating



**<1
YEAR**

COVID-19 vaccine development

**105
YEARS**

Typhoid fever (1884–1989)

Modern tools — mRNA technology, genome sequencing, and advanced microscopy — have collapsed development timelines from over a century to under a year.

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

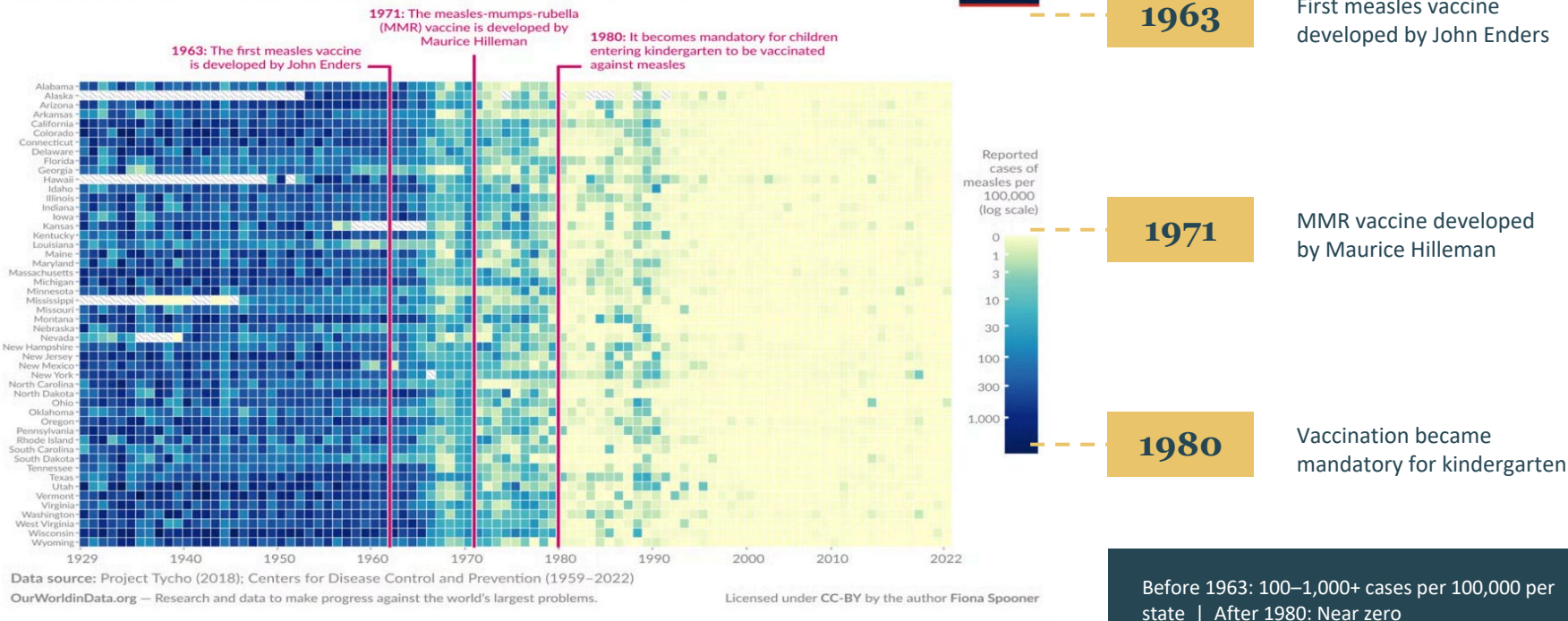
Pre-vaccine vs. post-vaccine annual case and death rates per million population | Source: Roush & Murphy (2007)

Disease	Pre-Vaccine Cases/million/yr	Post-Vaccine Cases/million/yr	Case Reduction	Pre-Vaccine Deaths/million/yr	Death Reduction
Diphtheria	158	0	100%	13.7	100%
Smallpox	250	0	100%	2.9	100%
Acute polio	141	0	100%	10	100%
Measles	3,044	0.2	99.99%	2.5	100%
Rubella	242	0.04	99.98%	0.09	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

 = 100% elimination of both cases and deaths

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Vaccines reduced measles cases across US states



Download image or data

Smallpox: The Only Human Disease Eradicated Through Vaccination

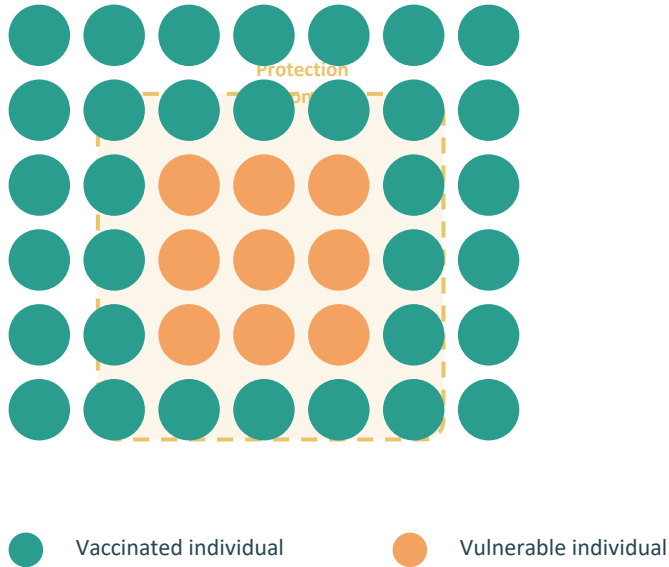
From WHO resolution to global certification in just 21 years



Why Smallpox Was Uniquely Suited for Eradication

- 01 The virus infected only humans — no animal reservoir to sustain transmission
- 02 Symptoms were easily distinguishable — enabling rapid identification and containment
- 03 The vaccine was cheap, stable, and effective — feasible for mass global distribution

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable



How Herd Immunity Works

When enough people are vaccinated, pathogens cannot spread effectively through a population. This creates a protective barrier that shields those who cannot be vaccinated.

Who relies on herd immunity:

Newborns

Too young to receive vaccinations

Immunocompromised

Weakened immune systems cannot mount response

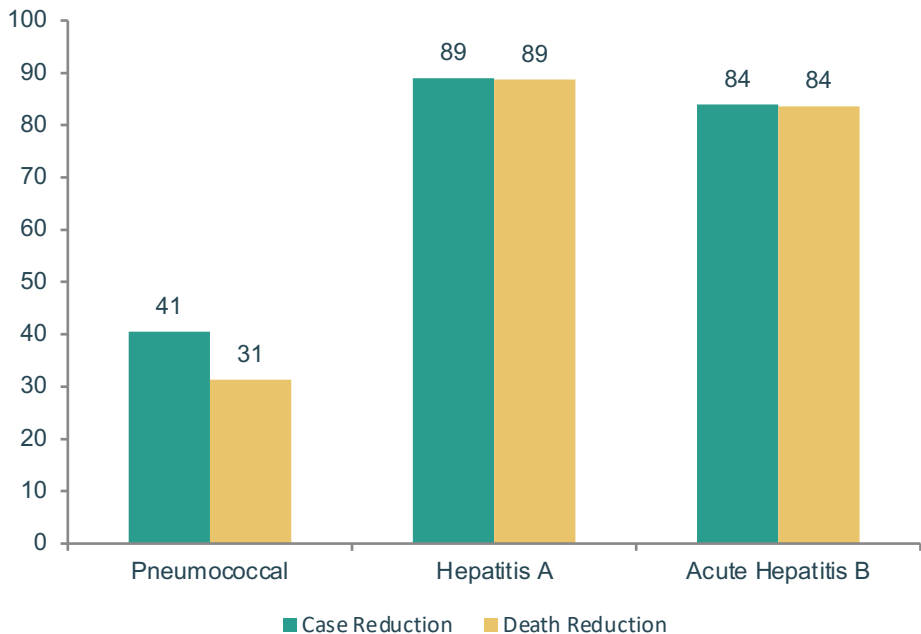
Chemotherapy patients

Cancer treatments suppress immunity

Vaccines protect not just individuals — they protect entire communities.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Diseases with incomplete vaccine impact in the US



100% = full elimination

Source: Roush & Murphy (2007), Our World in Data

Key Barriers to Full Coverage

Access Gaps

Globally, not everyone has access to vaccines — supply chains and infrastructure remain inadequate in many regions

Supply Shortages

Vaccine supply disruptions put children at risk, particularly in low-income countries

Trust Challenges

Poor scientific understanding and misinformation reduce vaccine confidence and uptake

Key Takeaways

Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

- 1 Vaccines achieved 100% elimination of cases and deaths for diphtheria, smallpox, and polio in the US
- 2 Development timelines collapsed from over 100 years (typhoid) to under 1 year (COVID-19), driven by mRNA technology and genomic tools
- 3 Measles cases dropped to near zero across all 50 US states following the 1980 kindergarten vaccination mandate
- 4 Smallpox remains the only human disease eradicated through vaccination, certified in 1980 after a global campaign
- 5 Gaps persist for pneumococcal disease (40.5% case reduction), hepatitis A (89%), and hepatitis B (83.9%)

Call to Action

Improve access, expand coverage, and strengthen trust in vaccination — global health systems must commit to closing the remaining gaps.